

Machine Learning Project 1

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Abstract - Coronary heart disease (CHD) is a major global health concern. This project aims to predict CHD risk using survey data from the 2015 BRFSS dataset. A complete machine learning pipeline was developed, combining systematic preprocessing, correlation-based feature selection, polynomial feature expansion, and regularized logistic regression with threshold tuning. The model achieved an F1-score of 0.433 on the AICrowd platform, performing well despite the dataset’s imbalance and noise.

1 Introduction

Coronary heart disease remains one of the leading causes of mortality worldwide. Large-scale health surveys such as the Behavioral Risk Factor Surveillance System (BRFSS) provide valuable population data but introduce challenges including missing values, mixed feature types, and severe class imbalance.

In this project, the focus is on predicting CHD using the BRFSS 2015 dataset with the MICHHD label as the binary indicator of disease presence. The goal was to design a transparent and reproducible baseline. Emphasis is placed on data quality, dimensionality reduction, and threshold tuning to handle imbalance, and it is shown that even simple models, when carefully engineered, can yield reliable predictions on large and noisy real-world datasets.

2 Methods

2.1 Data Preprocessing

After loading the dataset, numerous missing entries and special numeric codes representing missing or refused responses (e.g., 7, 8, 9, 77, 99, 999) were identified. These codes varied across columns and were first mapped to NaN. The first nine administrative columns and all features with more than 95% missing values were removed.

The remaining features were categorized as continuous, categorical, or binary. Binary variables encoded as {1, 2} were remapped to {1, 0}. Missing values were im-

puted using the mean for continuous features and the mode for categorical and binary ones. Continuous features were then standardized to zero mean and unit variance, while categorical and binary features were left unscaled. Finally, a constant offset (bias) column was added as the first feature to facilitate linear model fitting.

2.2 Feature Engineering

The dataset contained nearly 300 features, many of which were irrelevant or highly correlated. To improve model performance, several feature modification steps were applied. First, correlation-based feature selection was performed, retaining only features with a Pearson correlation to the target of at least 0.05 and removing one feature from each pair with inter-correlation above 0.95. This reduced redundancy and focused the model on the most informative predictors.

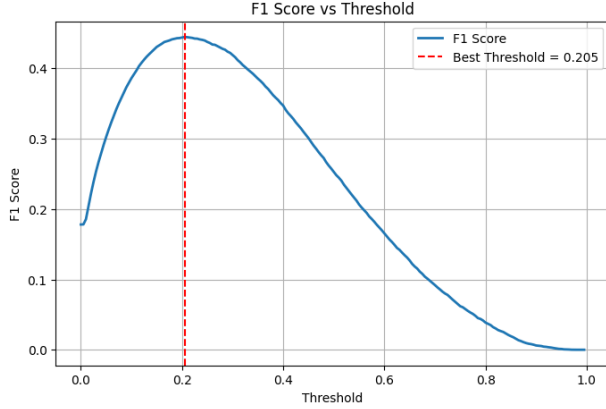
To capture nonlinear effects and interactions that linear models cannot express directly, a second-degree polynomial expansion was generated, including all squared and pairwise product terms, followed by standardization using the training statistics. Since this expansion produced approximately 2,500 features, correlation-based filtering was repeated with stricter thresholds ($|r| \geq 0.1$, inter-correlation ≤ 0.75), yielding a compact and expressive feature set that preserved predictive signal while maintaining computational efficiency.

2.3 Model Implementation

A regularized logistic regression model was trained and its performance evaluated using the cross-entropy loss. Since the dataset was highly imbalanced (approximately 90% negatives and 10% positives), majority-class undersampling was applied to obtain a more balanced training set. Because class imbalance affected prediction confidence, a threshold tuning procedure was implemented to improve classification performance.

After computing the predicted probabilities from the sigmoid output, the optimal threshold between 0 and 1 (in steps of 0.005) was selected by maximizing the F1-score on the training set. This threshold was then applied to the validation sets.

Model performance was assessed using the confusion matrix and derived metrics : accuracy, precision, recall, and F1-score. The results on the validation set confirmed that threshold tuning significantly improved the F1-score and provided a better balance between precision and recall, which is essential for imbalanced datasets.



(a) Dataset without feature engineering.

FIGURE 1 – Effect of threshold tuning on F1-score.

2.4 Model Selection and Validation

The regularization strength λ was selected using k -fold cross-validation ($k = 4$), with cross-entropy as the validation metric. For each $\lambda \in \{0, \dots, 0.1\}$ (15 values, linearly-spaced), a regularized logistic regression model was trained with fixed parameters `max_iters` = 100 and step size $\gamma = 0.01$.

In each cross-validation round, the model was trained on $k - 1$ folds and evaluated on the remaining fold. This process was repeated until every fold had served once as validation data, and the resulting validation losses were averaged to estimate the mean performance for that λ . The optimal regularization strength was then chosen as the one minimizing the mean validation loss.

Empirically, the validation loss was strictly increasing as λ grew, indicating underfitting at higher regularization levels. Consequently, $\lambda = 0$ yielded the best performance, and thus the final chosen model corresponds to a logistic regression model without regularization. This can be explained by the large number

of samples in the train set (238588 samples) and relative low complexity of the model (326 parameters), preventing overfitting.

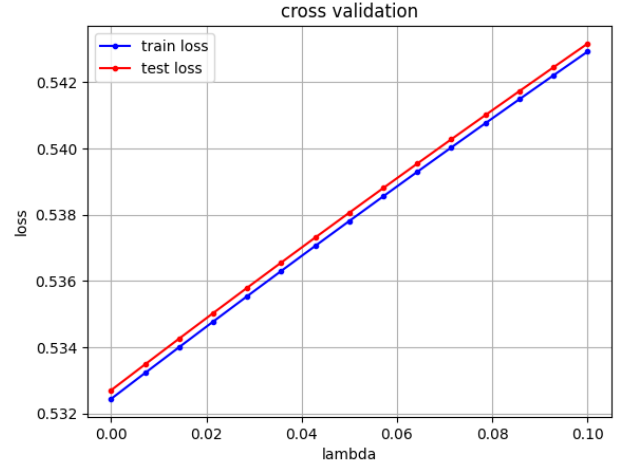


FIGURE 2 – Cross-validation curves showing train (blue) and validation (red) Loss as a function of λ .

3 Results and Conclusion

The final model achieved an **F1-score of 0.433** on the AICrowd platform, performing well compared to other teams. Although the dataset was large, noisy, and highly imbalanced (≈ 90 – 10), systematic preprocessing, correlation-based feature selection, and threshold tuning significantly improved performance and stability.

These results demonstrate that even a regularized logistic regression model, when carefully designed and tuned, can effectively handle complex real-world data and provide strong performance. In future work, this approach could serve as a baseline for comparison with more advanced models or automated feature selection methods.